

AirTag Teardown

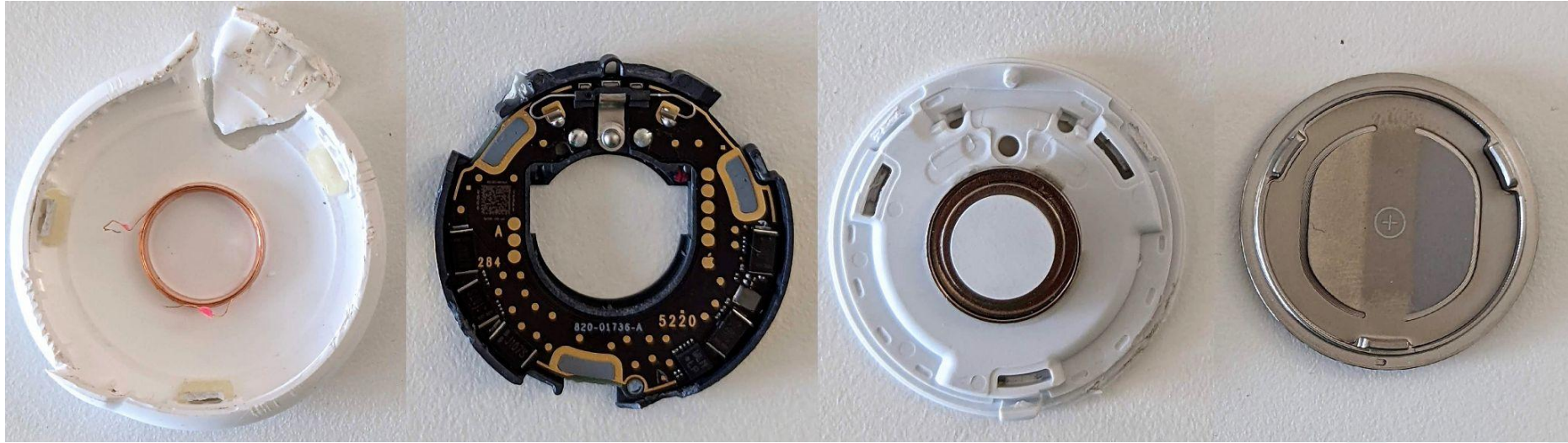
References :- <https://adamcatley.com/AirTag.html>
https://github.com/colinoflynn/airtag-recolinoflynn-testpoint_mapping



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Teardown



Key Findings



Hardware & Power

- Uses **off the shelf** components, apart from Apple's U1 chip for UWB
- The popular **nRF52832** is used for BLE and NFC
- Sleep current consumption of **2.3 μ A** gives over **10 years** of potential battery life
- nRF needs at least **~1.9V** battery voltage to boot up
- 32Mbit flash storage is **unencrypted**
- nRF programming interface (**SWD**) easily accessible via test pads and can be unlocked

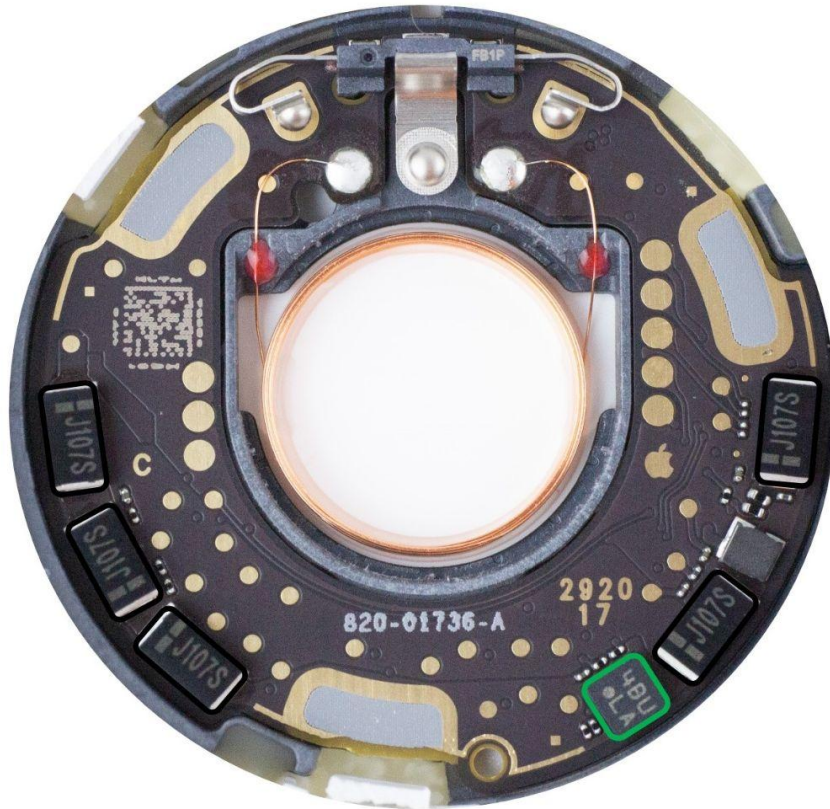
Connectivity & Operation

- Updates BLE address and public key **once a day** at 04:00am
- Updates last byte of advertisement data every **15 minutes**
- Transmits BLE advertisement every **2 seconds** when away from its owner's device
- BLE connection interval of **1 second** when near its owner's device

Lost Mode & Motion

- Goes into a lost mode exactly **3 days** after being away from owner
- Makes noise once every **6 hours** in lost mode when movement is detected
- Samples accelerometer every **10 seconds** when waiting for movement
- Samples accelerometer every **0.5 seconds** once motion is detected, for **20 seconds**

PCB Overview



Components used



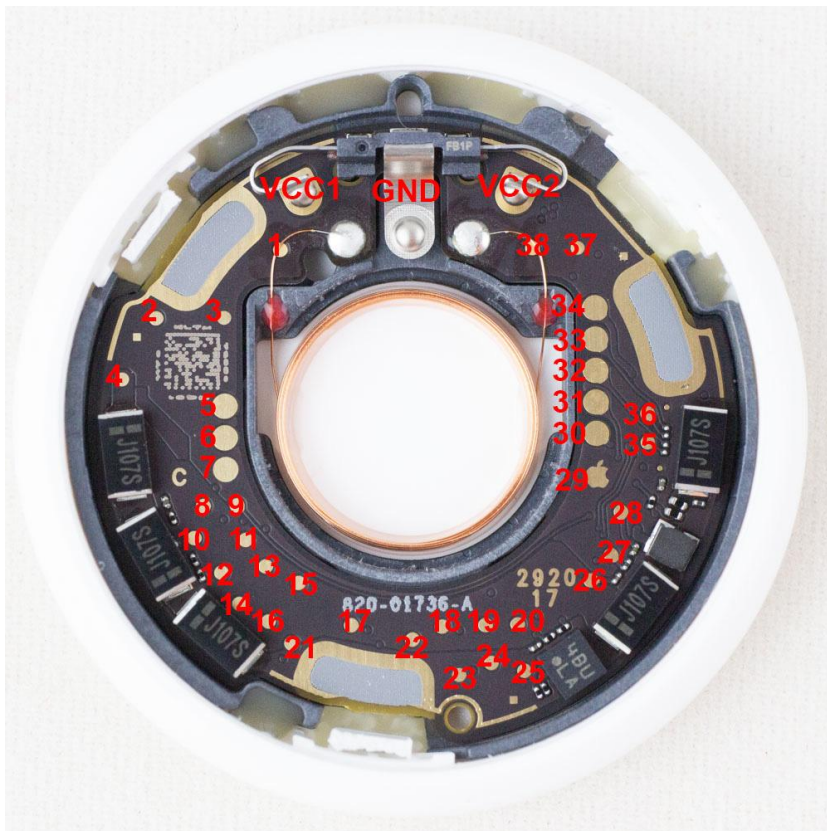
- Nordic nRF52832 SoC with BLE and NFC, plus 32MHz and 32.768kHz crystals
- Apple U1 UWB Transceiver
- GigaDevice GD25LE32D 32Mbit NOR flash
- Bosch BMA280 accelerometer
- Maxim MAX98357AEWL audio amplifier
- TI TPS62746 DC-DC buck converter
- TI TLV9001IDPWR opamp
- 100uF Electrolytic Capacitors (5x)
- Unknown. Unable to decode markings

Antennas



1. Bluetooth Low Energy (left) - 2.4GHz
2. NFC (middle) - 13.56MHz
3. Ultra-Wideband (right) - 6.5-8GHz

Test Points



Category	TP#	Type	Signal / Name	Description	Purpose / Notes
 Speaker	1	Speaker	SPK+ (Voice Coil 1)	One end of the voice coil	Drives the AirTag speaker (positive side)
	38	Speaker	SPK- (Voice Coil 2)	The other end of the voice coil	Completes the speaker circuit (negative side)
 Power	5	Power	VCC2	VCC2 in (3V)	Battery positive input 2 (≈3V)
	6	Power	VCC1	VCC1 in (3V)	Battery positive input 1 (≈3V)
	7	Power	GND	Ground	Main ground reference
	29	Power	GND (Apple Logo)	Ground near Apple logo	Additional ground point
 Flash	19	Flash	DATA IN (DI)	Data in	SPI Data input to Flash
	20	Flash	DATA OUT (DO)	Data out	SPI Data output from Flash
	21	Flash	VCC (1.8V)	1.8V supply for Flash	Power supply for the Flash chip
	22	Flash	SCLK	Serial Clock	SPI Clock signal
	23	Flash	CS (Chip Select)	Chip Select	Enables / selects the Flash chip
 nRF (MCU)	30	nRF	nRST	Reset	Hardware reset for nRF52832
 Debug	31	Debug	SWO	Serial Wire Output	Debug trace / output
	35	Debug	SWCLK	Serial Wire Clock	SWD Clock
	36	Debug	SWDIO	Serial Wire Data	SWD Data



iPhone 11



AirTag



U1 Architecture

- AirTag uses Apple's U1 Ultra-Wideband chip for precision ranging.
- Operates on UWB Channels 5 (6.5 GHz) and 9 (8 GHz) with 500 MHz bandwidth.
- Uses BPSK modulation and a single UWB antenna.
- The AirTag primarily performs accurate distance measurement using Time-of-Flight (ToF).
- Direction estimation is likely computed by the iPhone using multiple antennas and spatial processing techniques such as AoA/AoD.
- This architecture enables compact size, low power consumption, and high ranging accuracy.

Higher Resolution Images

